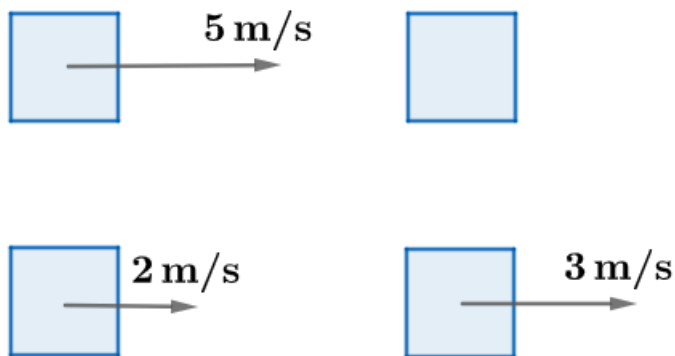


2022A F=ma Exam: Problem 3

Kevin S. Huang



By conservation of energy, the first block continues moving in its initial direction. Conserving linear momentum,

$$\begin{aligned} m(5 \text{ m/s}) &= m(2 \text{ m/s}) + mv_f \\ v_f &= 3 \text{ m/s} \end{aligned}$$

Thus, the initial kinetic energy is

$$K_0 = \frac{1}{2}m(5 \text{ m/s})^2$$

The final kinetic energy is

$$K_f = \frac{1}{2}m(2 \text{ m/s})^2 + \frac{1}{2}m(3 \text{ m/s})^2$$

The fraction lost is

$$f = \frac{K_0 - K_f}{K_0} = 1 - \frac{K_f}{K_0} = 1 - \frac{2^2 + 3^2}{5^2} = 0.48$$

so the answer is C.